

Week 1.2 — Soldering Bootcamp

June 19 – 21, 2026 · Three sub-sessions · Fri / Sat / Sun

Lesson 1.2 was soldering bootcamp — the team's first weekend with hot irons in their hands. It ran as three sub-sessions across three days because workstations were the bottleneck and schedules were uneven. Before any of the new kids touched a kit, Coach John ran Trenton — who had prior soldering experience — through a full test build on Thursday June 18. That prep session is where the four soldering rules below actually came from. Friday June 19 brought the first live cohort (Rebecca, Isaac, Liam, Maddox, Tyler) through the full sequence: safety brief, general instruction, then practice board on the SpaceWings DIY Lucky Wheel (NE555 + CD4017 spinner). About an hour per board; iron at 650°F. Saturday June 20 was a two-person makeup (Evan + Rhylan) who split one board between them. Sunday June 21, 3:00–5:15 PM, was the full Sunday cohort (Maddox, Frankie, Trenton, Adrian, Isaac, Liam, Luke). Sunday added a new block: splicing two wires, waterproofing the joint with hot glue and heatshrink, then dunking in salt water to test. Nine joints built. Two passed. Seven leaked. The waterproofing-test section below has the method, the failure analysis, and the four fixes that come out of a 22% first-try pass rate — which is exactly what early-season failure is for.

3 SUB-SESSIONS	11/12 UNIQUE STUDENTS	~1 hr PER BOARD	2 / 9 WATERPROOF JOINTS
SAFETY - Solder is lead-based. Don't inhale the fumes; wash hands when you're done. - Place a fan to extract or blow fumes away from the student. Direction matters — away, not toward. - The iron tip burns. So do the back of the board and the component leads. Treat both as hot. - Safety glasses on whenever you're snipping leads. Small metal clippings fly off the cutters.		DECISIONS MADE - Coach John runs the safety brief and the general soldering instruction at each session. - Trenton handles pre-class test builds on new kits, to surface lessons before the rest of the team. - Iron temperature standardized at 650°F for the team's practice kits. - Split the class across multiple sessions when workstations or schedules are the bottleneck. - Splitting one practice board between two students works as a fallback when boards are short. - Salt-water + multimeter is the team's standard waterproof test. Infinite resistance = sealed. - Safety brief at the start of every soldering session. No exceptions.	
WHAT WE LEARNED - Work from the inside of the board outward. Anything you've soldered is a vertical obstacle to whatever comes next; building outward keeps your iron's path open. - Small components before big ones. Resistors and low-profile caps first; headers and tall connectors last. Don't build a wall you have to reach over. - Right-handed solderers do multi-pin parts left to right. Left-handed, right to left. Otherwise your hand drags across joints you just made. - Solder the battery pack absolute last. The pack hangs off the PCB and blocks access to every joint underneath it.		ACTION ITEMS Rebuild the 7 failed waterproof joints with the new technique; re-run the salt-water test <i>Sunday cohort · by 1.3</i> Bring kits and components to next class (Lesson 1.3) <i>Advisor · 6/28</i> Set the date for officer elections (deferred from 1.1) <i>Advisor · by 1.3</i> Practice soldering at home if a station is available <i>Every student · ongoing</i> Add Sat + Sun session photos to the photo log once parent releases are returned <i>Photographer · when consent is in</i>	
STUDENT LEARNING OUTCOMES - Importance of polarity in certain components — diodes, electrolytic caps, transistors — and the habit of slowing down to check before committing the joint. - Soldering technique, equipment care, and safety habits around hot tools and lead-based material. Including the visual + tactile difference between a cold joint and a good one. - Waterproofing is iteration. The 2-of-9 first-try pass rate isn't a failure — it's how you find where the real seal points are. The four fixes become the standard procedure going forward.		WATERPROOFING TEST - Method: 2 cups warm water + 2 tbsp sea salt. Multimeter on 2000kΩ. One probe in the water, one on the open end of the wire. Submerge the joint. Infinite resistance = sealed; finite = leak. - Result on first try: 2 of 9 joints passed. 22%. Most leaked. - Root cause: water seeped in through improperly heatshrunk wires — wires poking out, heatshrink not tight, seal not continuous. - Fix 1 — Use longer heatshrinks so the seal extends past the joint on both sides. - Fix 2 — Hot-glue underneath the heatshrink before shrinking. The glue fills voids the heatshrink can't. - Fix 3 — Clip the poking wires flush before sealing. A poking wire is a leak path. - Fix 4 — If the heatshrink is too big, layer a smaller one underneath to tighten the seal.	
LOOKING AHEAD <i>Class 1.3 — Kit Unboxing and First Electronics — is Sunday June 28. Kits get opened, components get identified, and we move from practice joints to circuits that actually do something. The Sunday cohort also owes a rebuild of the 7 failed waterproof joints with the new technique and a re-run of the salt-water test. Closed-toe shoes still required.</i>			

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